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WEARABLE SEATING ASSEMBLY

Abstract

A wearable seating assembly is provided. The wearable seating assembly includes a seat and a strap. The body of the seat is secured to the body of the user with the strap, which is threaded through two slots in the seat. The slots are above the middle of the seat, placing most of the body of the seat below the hips of the user. When the user sits, the seat bends under the user providing a clean and dry surface to sit on, without any manual manipulation of the seat or strap by the user. When the user stands, the seat returns to its original position, again without any manual manipulation. The user may alter how the straps pass through the slots to alter how the seat rests against the body of the user. The seat may have warming elements to make the seat warm. The seat may include a safety strap to prevent the seat being lost.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates generally to a wearable seat.

BACKGROUND OF THE INVENTION

[0002] There is a general desire when in cold or wet environments to have easy access to seating that is warm and dry, particularly when for example participating in outdoor activities such as hiking, skiing, snowboarding, fishing, hunting, and other instances where sitting on a hard, cold and/or damp surface can be uncomfortable. Similar issues may be experienced in certain work settings such as construction and resource extraction.

[0003] Carrying a seat is often impractical in recreational or work settings. There is therefore a desire to have a seat that can be transported without needing to be carried independently.

[0004] Seats that can be “worn” by a user are known. Prior art wearable seats require users to modify or otherwise interact with the seat to use it. This makes the seats more difficult and time consuming to use, and requires a user to have free hands to use.

[0005] The present invention addresses these issues.

[0006] The foregoing examples of the related art and limitations related thereto are intended to be illustrative and not exclusive. Other limitations of the related art will become apparent to those of skill in the art upon a reading of the specification and a study of the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Exemplary embodiments are illustrated in referenced figures of the drawings. It is intended that the embodiments and figures disclosed herein are to be considered illustrative rather than restrictive.

[0008] FIG. 1 shows an embodiment of the seat of a wearable seat.

[0009] FIG. 2 shows the embodiment of FIG. 1 with additional reference datum.

[0010] FIG. 3 shows the first seating face of the embodiment of FIG. 1.

[0011] FIG. 4 shows an embodiment of a wearable seating assembly with the strap in a first configuration.

[0012] FIG. 5 shows an embodiment of a wearable seating assembly with the strap in a second configuration.

[0013] FIG. 6a shows a cross section of another embodiment of the seat of a wearable seat, showing heating element pockets.

[0014] FIG. 6b shows a cross section of another embodiment of the seat of a wearable seat, with a single heating element pocket.

[0015] FIG. 6c shows a cross section of another embodiment of the seat of a wearable seat, with a heating element pockets where the seating element slot is the same dimension of the seating element pocket.

[0016] FIG. 7 shows a user wearing an embodiment of a wearable seating assembly when walking or standing.

[0017] FIG. 8 shows a user wearing an embodiment of a wearable seating assembly when sitting.

DETAILED DESCRIPTION OF THE INVENTION

[0018] Throughout the following description specific details are set forth in order to provide a more thorough understanding to persons skilled in the art. However, well known elements may not have been shown or described in detail to avoid unnecessarily obscuring the disclosure. Accordingly, the description and drawings are to be regarded in an illustrative, rather than a restrictive, sense.

[0019] It should be noted that for ease of description, the drawings merely show the related parts of the present application. In a case of no conflict, the embodiments of the present application and features in the embodiments may be combined with each other.

[0020] The term “generally” as used herein refers to a characteristic, feature, or aspect that is predominantly, but not necessarily exclusively, present or true. It implies a degree of flexibility or variation, allowing for instances or embodiments where the characteristic, feature, or aspect may deviate from a strict interpretation. This term is used to describe a typical or common condition, shape, or arrangement, acknowledging that alternative forms or variations might exist within the scope of the invention.”

[0021] The terms “a”, “an”, and/or “the” as used herein do not specifically refer to singular, but may also include plural, unless the context clearly indicates exceptions. Generally speaking, the terms “include” and “comprise” only indicate that the features that have been clearly identified are included, and these steps and features do not constitute an exclusive list. Devices may also contain other features. A feature defined by the statement “comprising (including) a . . .” does not exclude the case that other identical features exist in the product or device including the feature.

[0022] The terms “first” and “second” as used herein are for purpose of description, and should not be interpreted as indicating or implying relative importance or implying the number of the indicated technical features. Therefore, the features defined by “first” and “second” may explicitly or implicitly include one or more of the features.

[0023] Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. Where a specific range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is included therein. All smaller sub ranges are also included. The upper and lower limits of these smaller ranges are also included therein, subject to any specifically excluded limit in the stated range.

[0024] FIG. 1 shows a seat **103** of a wearable seating assembly **101** according to an embodiment. In some embodiments seat **103** is of unitary construction. In other embodiments seat **103** may be constructed of two or more separate pieces. Seat **103** has a generally planar body **105**. In this non-limiting embodiment body **105** is generally a rectangular shape. Those skilled in the art will recognize that body **105** may be any shape or dimensions suitable for the purposes described herein. For example, in some embodiments seat **103** may be heart-shaped or semicircular. In some embodiments, seat **103** may have dimensions intended for specific demographics. For example a wearable seating assembly for adults may have height **123** between 25 cm and 35 cm, length **125** between about 35 cm to 50 cm, and thickness **124** between about 1.3 cm and 7 cm; or a wearable seating assembly for children may have height **123** between about 10 cm and 18 cm and length **125** between about 14 cm to 25 cm, and thickness **124** between about 1.3 cm and 7 cm.

[0025] Body **105** may be comprised of a single material or multiple materials. In some embodiments, the material of body **105** is semi-rigid, has low thermal conductivity, is non-water-permeable, and/or has a low density (to reduce weight and enhance cushioning). In some embodiments, the material of body **105** has a thermal resistance between 0.1761

[00001] $\frac{K \cdot \text{Math. } m^2}{W}$
and 0.8805

[00002] $\frac{K \cdot \text{Math. } m^2}{W}$.

In some embodiments, body **105** weighs between 50 g and 250 g. An example of such a material for body **105** is closed cell polyethylene foam.

[0026] Seat **103** has a first seat face **107** and a second seat face **109** opposite first seat face **107**. When wearable seating assembly **101** is being worn by a user, first seat face **107** abuts the body of

the user and second seat face **109** faces away from the body of the user. Seat **103** further has a top face **111** and a bottom face **113** opposite top face **111**. When wearable seating assembly **101** is being worn by a user top face **111** is closer to the user's head than bottom face **113**. Each of first seating face **107**, second seating face **109**, top face **111**, and bottom face **113** are comprised of one or more surfaces of body **105**.

[0027] Seat **103** further includes a first slot **115** and a second slot **117**, each extending through body **105** from first seat face **107** to second seat face **109**.

[0028] In some embodiments, seat **103** has one or more thinning elements to decrease the vertical rigidity of seat **103** in the region around the thinning elements. The non-limiting embodiment shown in FIG. **1** has one thinning element comprising seam **119**. Seam **119** comprises a cut through the material of body **105** from first seat face **107** towards second seat face **109**. In some embodiments seam **119** comprises a cut through the material of body **105** from second seat face **109** towards first seat face **107**. In some embodiments seam **119** comprises a cut through the material of body **105** from both second seat face **109** towards first seat face **107** and from first seat face **107** towards second seat face **109**. In some embodiments, seam **109** has a depth between 0.3 and 0.75 cm. In some embodiments seam **109** has no thickness (i.e. is formed by cutting the material without removing material). In some embodiments seam **109** has a thickness (i.e. is formed by removing material). In some embodiments the thickness of seam **109** is between 0.1 and 1 cm. Other thinning methods may include: removal of material from one or both of first seat face **107** and second seat face **109**; or sewing between first seat face **107** and second seat face **109**.

[0029] In some embodiments, thinning element reduces the distance between first seat face **107** and second seat face **109** by a minimum of 10% and a maximum of 95% of the original distance between first seat face **107** and second seat face **109**. In some embodiments, a thinning element is located between 0.5 and 5 cm below first slot **115** and second slot **117**. In some embodiments, thinning elements are located between 20% and 80% of the distance between top face **111** and bottom face **113**.

[0030] In some embodiments, bottom face **113** of seat **103** incorporates a taper **121** such that first seat face **107** extends lower than second seat face **109**. Taper **121** may extend fully (as illustrated in the FIGS. **1** and **2**) or partially from first seat face **107** to second seat face **109**. In some embodiments, taper **121** has an angle relative to first face **107** of no greater than 45° from the horizontal.

[0031] FIG. **2** shows seat **103** with additional reference elements. Seat **103** has a middle point **201**. The location of middle point **201** is derived using an arithmetic means. In this example embodiment, middle point **201** is at the center of mass of the body **105**. Other non-limiting examples of means of deriving the location of middle point **201** include placing the middle point **201** at the centroid of the body **105**, or at a point equidistance from the extremities of the body **105** along three perpendicular axes. First slot **115** and second slot **117** are equidistantly and opposingly spaced from middle point **201**.

[0032] Middle point **201** is on a horizontal reference plane **203**. The intersection of horizontal reference plane **203** with body **105** is shown by line **205**. A vertical reference plane **207** is normal to horizontal reference plane **203**. The intersection of vertical reference plane **207** with body **105** is shown by line **208**. In this embodiment middle point **201** is on vertical reference plane **207** but this is not required in all embodiments. Vertical reference plane **207** divides the body **105** into a first section **209** and a second section **211**.

[0033] A cross section reference plane **213** is normal to both vertical reference plane **207** and horizontal reference plane **203**. In this non-limiting embodiment cross section reference plane **213** is co-planar with the second seating face **109**. This is not required and may be impossible in embodiments where the second seating face **109** is not planar. Cross section reference plane **213** may be used to generate cross sections of body **105**, such as cross section **215**. Cross section **215** intersects body **105** at line **217**.

[0034] Cross sections such as cross section **215** can be used in the arithmetic derivation of middle point **201**. An example of a derivation is placing middle point **201** at the centroid of a cross-section of body **105**. Another derivation involves drawing a first line parallel with horizontal reference plane **203** so that first line divides the cross section of body **105** into portions of equal area, then drawing a second line parallel with vertical reference plane **207** so that the second line divides the cross section of the body **105** into portions of equal area, then placing middle point **201** at the intersection of the first and second lines.

[0035] A slot reference plane **219** is parallel with horizontal reference plane **203**, and positioned between horizontal reference plane **203** and top face **111**. The intersection of slot reference plane **219** with body **105** is shown by line **221**.

[0036] FIG. **3** shows seat **103**, as seen looking orthogonally to first seat face **107**. This view aids in showing how the first slot **115** and the second slot **117** are positioned. Line **208** shows the reflection of vertical reference plane **207** onto body **105**, which divides body **105** into first section **209** and second section **211**. The volume of first slot **115** must be entirely within first section **209** and the volume of second slot **117** must be entirely within second section **211**. More than 50% of the volume of each of first slot **115** and second slot **117** must be above horizontal reference plane **203** (shown reflected onto body **105** by line **205**). In this non-limiting embodiment, 100% of the volume of first slot **115** and second slot **117** are above horizontal reference plane **203**. At any cross section taken parallel to cross section reference plane **213**, slot reference plane **219** must pass through the profiles of first slot **115** and second slot **117**.

[0037] FIG. **4** shows wearable seating assembly **101** with both seat **103** (which has been previously described) and a strap **401**. Strap **401** threads through first slot **115** and second slot **117** and thus is upwardly offset from a middle point **201** of seat **101**. The term “upwardly offset” as used herein means an upward offset of a minimum of 15% and a maximum of 65% of the distance between middle point **201** and top face **111**. For example, in embodiments designed for use by adults the upward offset may be between 2 cm and 13 cm; or in embodiments designed for use by children the upward offset may be between 1 cm and 7 cm.

[0038] Strap **401** has a mesh **403** and may include an adjustment mechanism **405**. Mesh **403** can be made partly or wholly of an elastic material, such as spandex. An adjustment mechanism **405** allows strap **401** to form a closed loop **407**, and can adjust the circumference of closed loop **407**.

[0039] Appropriate adjustment mechanisms are known to the art and will not be described in detail here. Depending on the type of the adjustment mechanism **405**, closed loop **407** may be disconnected. Adjustment mechanisms allowing disconnection include side release buckles, double d-ring buckle, and hook-and-loop fastener. Examples of adjustment mechanisms that do not allow disconnection include tension locks, cam locks, and slide adjusters.

[0040] Wearable seating assembly **101** has two configurations of strap **401**, defined by the positioning of mesh **403** relative to first slot **115** and second slot **117**. FIG. **4** shows wearable seating assembly **101** in a first configuration. In the first configuration, wherein strap **401** forms closed loop **407**, starting at adjustment mechanism **405** and going along mesh **403**: mesh **403** goes through first slot **115** from second seat face **109** and exits first slot **115** on first seat face **107**, then mesh **403** enters second slot **117** from first seat face **107** and exits second slot **117** onto second seat face **109** before returning to adjustment mechanism **405**.

[0041] FIG. **5** shows wearable seating assembly **101** in a second configuration. In the second configuration, using the same notation as previously, mesh **403** enters first slot **115** from first seat face **107** and exits first slot **115** onto second seat face **109**, then mesh **403** enters second slot **117** onto the second seat face **109** and exits second slot **117** onto first seat face **107**.

[0042] Wearable seating assembly **101** can change between the first configuration and the second configuration. The first configuration provides a relatively snugger fit of seat **101** against the user since strap **401** bends outer side sections of seat **101** against the user's body, especially as strap **401** is tightened. In the second configuration seat **101** maintains its original planar shape.

[0043] A method of changing configuration involves rotating the seat **103** approximately 180 degrees around a central longitudinal axis **501**. Another method of changing configuration involves a user removing strap **401** from seat **103** then replacing strap **401** in the other configuration. The method of changing between the first configuration and the second configuration may depend on the particular embodiment of wearable seating assembly **101**.

[0044] One skilled in the art will see advantages to either method of changing configurations. The former requires minimal effort from the user to change between configurations but will change first seat face **107** into second seat face **109** and vice versa. This may be disadvantageous if, for example, the face becoming first seat face **107** is wet or dirty, which would then be transferred onto the user when the seat is used. The later method does not change first seat face **107** and second seat face **109** but requires greater time and effort by the user to change between configurations.

[0045] FIG. **6a** shows a cross section of wearable seating assembly **101** along cross section **215**. Wearable seating assembly **101** may include one or more heating element pockets **601** formed in an interior of seat **103**. A user may place removable heating elements into heating element pockets **601** to make first seating face **107** warm to improve user comfort in certain conditions. Embodiments with heating element pockets **601** may be used with or without the removable hearing elements. Removable heating elements may comprise any portable heat source, including but not limited to chemical heat sources and electrical heat sources. The non-limiting embodiment shown in FIG. **6a** includes chemical heat sources in the form of disposable chemical hand warmers (such as HotHands™ brand hand warmers). Electrical heat sources can include but are not limited to electrical hand warmers or dedicated heating elements. Wearable seating assembly **101** having one or more heating element pockets **601** may or may not include thinning element such as seam **119**. In embodiments with one or more heating element pockets **601**, seam **119** does not penetrate through the material of body **105** to heating element pockets **601**.

[0046] As seen in FIGS. **1** and **6**, removable heating elements can be inserted or removed from seat **103** through heating element slots **603**. As shown in FIG. **6a**, heating element slots **603** connects to heating element pockets **601**. In this non-limiting embodiment two heating element pockets **601** of a generally rectangular shape are provided in top and bottom arrangement. Heating element pockets **601** can be in any number, shape and arrangement. In this non-limiting embodiment heating element slot **603** are in top face **111** and bottom face **113**.

[0047] FIG. **6b** shows a non-limiting embodiment with a single heating element pocket **601b**. In other embodiments, heating element slots **603** may be anywhere on body **105** that allows access to heating element pockets **601**.

[0048] As seen in FIG. **6c**, in some embodiments heating element pocket **601** may have the same profile as heating element slot **603**. In this non-limiting embodiment heating element pocket **601** extends approximately 80% of the distance between top face **111** and bottom face **113**.

[0049] In some embodiments the material of body **105** is elastically deformable. In these embodiments heating element pockets **601** may be slightly smaller than the removable heating elements and deform to accommodate removable heating elements, firmly securing them.

[0050] In some embodiments heating element pocket(s) **601** may simply be a hollowed out interior portion of seat **103** of dimensions that do not necessarily correspond to the dimensions of heating elements. In other embodiments the dimensions of heating pocket(s) **601** do correspond to the dimensions of heating elements.

[0051] In some embodiments, body **105** may have features to improve the comfort of users when using the wearable seating assembly **101** with heating elements. For example, users are generally more comfortable when: a greater amount of the thermal energy released from the heating elements is radiated from first seat face **107** (i.e. towards the user) than from second seat face **109** (i.e. away from the user); and when the thermal energy from the removable heating elements is disbursed over a great area of first seat face **107** (assists in avoiding overheating or burns on portions of users in contact with first seat face **107**). Embodiments may include one or more feature to achieve either

or both of these functions.

[0052] In some embodiments, body **105** may have greater thermal resistance between heating element pockets **601** and second seating face **109** than between heating element pockets **601** and first seating face **109**. Non-limiting methods of achieving this include: [0053] constructing body **105** with a material with lower thermal conductivity between heating element pockets **601** and second seating face **109**; [0054] constructing body **105** with a layer of material with high thermal resistance between heating element pockets **601** and second seating face **109**; [0055] constructing body **105** with a layer of reflective material (such as reflective foil) between heating element pockets **601** and second seating face **109** to reflect heat in the direction of the user's body; [0056] partially or fully perforating body **105** between first seat face **107** and heating element pockets **601**; and/or [0057] positioning heating element pockets **601** closer to first seat face **107** than to second seat face **109**.

[0058] In some embodiments the material of body **105** is selected to have a low thermal conductivity to increase the area over which thermal energy from the heating elements is radiated on the first seat face **107**. In some embodiments, body **105** may include one or more thermal radiating elements between heating element pockets **601** and first seat face **107**. The thermal radiating elements may have a different thermal conductivity from other materials in the body **105**, and diffuse thermal energy generated by the removable heat sources across part or all of first seat face **107**.

[0059] In some embodiments, one or more “placeholder” foam pieces may be inserted into heating element pockets **601** to maintain the shape of heating element pockets **601** when not in use. In other embodiments, heating element pockets **601**, heating element slots **603**, and the entrance to heating element slots **603** may be airtight when the entrance is closed, so that heating element pockets **601** do not collapse when a user sits on seat **103**.

[0060] FIG. 7 shows an embodiment of wearable seating assembly **101** being worn by a user while standing or walking. Wearable seating assembly **101** includes a safety strap **701**. Safety strap **701** prevents wearable seating assembly **101** from being lost in the event of strap **401** failing or breaking. This may be useful, for example, when outdoors where loss may go unnoticed or on work sites where loss may create a hazard. Safety strap **701** may also be used to secure wearable seating assembly **101** to other items worn or carried if the user wishes to take off wearable seating assembly **101** by unbuckling strap **401**. Since seat **101** is waterproof it need not be stored and occupy space in for example a backpack. In some embodiments, safety strap **701** may attach one or more “placeholder” foam pieces when they are not in use. These examples are non-limiting and are for illustrative purpose only.

[0061] Safety strap **701** is attached to wearable seating assembly **101** at a connection point **703**. Connection point **703** may be adjacent top face **1** at upper first seat face **107** and second seat face **109**. Methods of connecting straps to an object are known to art. Non-limiting examples may include sewing an end of a strap or attaching the strap to a rivet with a fastener. End **705** of safety strap **701** has an attachment device **707** which is attached to the user at an attachment point **709**. Attachment devices are known in the art and will not be described here. Non-limiting examples of attachment devices include carabineers, d-rings, and hook and loop fasteners. Attachment point **709** can be on anything worn or carried by the user, including but not limited to belt loops, backpacks, or clothing.

[0062] Strap **401** attaches at or around the hips of the user. Strap **401** holds wearable seating assembly **101** against the lower back/buttocks of the user. This positioning, in addition to the lightweight nature and shape of wearable seating assembly **101**, prevents wearable seating assembly **101** from interfering with the user walking or running, and wearing, carrying, or using other garments, bags, or equipment. The user can thus move about with minimal impairment caused by using wearable seating assembly **101**. The position of first slot **115** and second slot **117** above the horizontal reference plane **203** places most of the area of seat **103** below the hips of the

user.

[0063] FIG. 8 shows the user wearing wearable seating assembly **101** in a sitting position. When the user sits, or leans so the buttocks of the user would otherwise contact an external surface, a lower portion of seat **103** initially contacts the external surface. As the user's weight presses second face **109** of seat **103** against the external surface, seat **103** deforms, bending to provide a seating surface between the user and the external surface underneath. Slot **119** and/or taper **121** can be employed in some embodiments to encourage seat **103** to fold under the user easily regardless of the angle at which the user approaches a sitting surface. When the user returns to a standing position, seat **103** returns to the shape shown in FIG. 7. Importantly, the user does not need to manually manipulate any part of wearable seating assembly **101** to change between sitting and standing/walking positions, i.e., wearable seating assembly **101** provides hands-free operation throughout, and the position of strap **401** returns to the user's hips when the user returns to a standing position.

[0064] In some embodiments, when initially sitting down, the user starts bending down slightly in front of the sitting surface. When the user is sufficiently low (i.e., a lower portion of seat **101** makes contact with the sitting surface) the user continues to bend down, but also incorporate a backwards motion. This causes seat **101** to bend (as opposed to possibly translating upwards). Initially seat **101** will only bend, but as it reaches the end capacity of its bending and the user is almost fully seated, the user may optionally pull the foam seat underneath the user, causing seat **101** to slightly slide further down the user hips as permitted by the elasticity of strap **401**.

[0065] Where a component (e.g. seat, strap, pocket, slot, mesh, etc.) is referred to above, unless otherwise indicated, reference to that component should be interpreted as including as equivalents of that component any component which performs the function of the described component (i.e. that is functionally equivalent), including components which are not structurally equivalent to the disclosed structure which performs the function in the illustrated exemplary embodiments of the invention.

[0066] While a number of exemplary aspects and embodiments have been discussed above, those of skill in the art will recognize certain modifications, permutations, additions and sub-combinations thereof. For example, in some embodiments: [0067] the strap may be two straps, each attached to respective sides of the seat, i.e, the seat does not have slots for threading the strap therethrough; [0068] the strap may be a continuous loop of elastic mesh instead of having an adjustment mechanism; [0069] the thinning element may be absent; [0070] the seat may comprise multiple different forms of the thinning element; [0071] the seat may have multiple safety straps to simultaneously attach to a user and to “placeholder” foam pieces; [0072] the heating element pocket may be absent; [0073] the safety strap may be absent; and [0074] the second seat face may be printed with various designs and text, such as for branding, marketing or custom personalization.

[0075] Specific examples of apparatus and methods have been described herein for purposes of illustration. These are only examples. The technology provided herein can be applied to systems other than the example systems described above. Many alterations, modifications, additions, omissions, and permutations are possible within the practice of this invention. This invention includes variations on described embodiments that would be apparent to the skilled addressee, including variations obtained by: replacing features, elements and/or acts with equivalent features, elements and/or acts; mixing and matching of features, elements and/or acts from different embodiments; combining features, elements and/or acts from embodiments as described herein with features, elements and/or acts of other technology; and/or omitting combining features, elements and/or acts from described embodiments.

[0076] Various features are described herein as being present in “some embodiments”. Such features are not mandatory and may not be present in all embodiments. Embodiments of the invention may include zero, any one or any combination of two or more of such features. All

possible combinations of such features are contemplated by this disclosure even where such features are shown in different drawings and/or described in different sections or paragraphs. This is limited only to the extent that certain ones of such features are incompatible with other ones of such features in the sense that it would be impossible for a person of ordinary skill in the art to construct a practical embodiment that combines such incompatible features. Consequently, the description that “some embodiments” possess feature A and “some embodiments” possess feature B should be interpreted as an express indication that the inventors also contemplate embodiments which combine features A and B (unless the description states otherwise or features A and B are fundamentally incompatible).

[0077] It is therefore intended that the following appended claims and claims hereafter introduced are interpreted to include all such modifications, permutations, additions, omissions, and sub-combinations as may reasonably be inferred. The scope of the claims should not be limited by the preferred embodiments set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole.

Claims

1. A wearable seating assembly comprising: a. a generally planar seat, wherein the seat comprises a first seat face for abutting against a user and a second seat face for contacting a surface on which the user intends to sit; and b. a strap connected to the seat, the strap upwardly offset from a middle point of the seat.
2. A wearable seating assembly according to claim 1 wherein the seat comprises a first slot and a second slot, wherein the first slot and the second slot are equidistantly and opposingly spaced from the middle point.
3. A wearable seating assembly according to claim 2 wherein the strap threads through the first slot and the second slot to form a closed loop.
4. A wearable seating assembly according to claim 3 wherein the strap is elastic or an elastic mesh.
5. A wearable seating assembly according to claim 4 wherein the strap comprises an adjustment mechanism for adjusting a circumference of the closed loop.
6. A wearable seating assembly according to claim 2 wherein: in a first configuration the strap enters the first slot through the second seat face and exits the first slot out the first seat face, and enters the second slot through the first seat face and exits the second slot out the second seat face; and in a second configuration the strap enters the first slot through the first seat face and exits the first slot out the second seat face, and enters the second slot through the second seat face and exits the second slot out the first seat face.
7. A wearable seating assembly according to claim 1 wherein the seat comprises at least one heating element pocket for holding a heating element in an interior of the seat, and a heating element slot for accessing the at least one heating element pocket, wherein the at least one heating element pocket is dimensioned smaller than the heating element, whereby the interior of the seat is deformable to accommodate the heating element.
8. A wearable seating assembly according to claim 7 where the seat comprises additional insulating material between the at least one heating element pocket and the second seat face.
9. A wearable seating assembly according to claim 7 wherein the seat comprises thermal radiating material between the at least one heating element pocket and the first seat face, wherein the thermal radiating material is configured to diffuse thermal energy generated by the heating element across part or all of the first seat face, wherein the heating element is a chemical heat source.
10. A wearable seating assembly according to claim 1 further comprising a safety strap, a first end of the safety strap attached to an attachment point of the wearable seat, and a second end of the safety strap configured to attach to an external attachment point on a user and not on the wearable seat, wherein the attachment point is on a top face of the seat.

- 11.** A wearable seating assembly according to claim 1 wherein the seat is generally constructed of a material that is semi-rigid, low density, waterproof, and has low thermal conductivity.
- 12.** A wearable seating assembly according to claim 11 wherein the seat is generally constructed of closed cell polyethylene foam.
- 13.** A wearable seating assembly according to claim 1 wherein the seat further comprises one or more thinning elements.
- 14.** A wearable seating assembly according to claim 13 wherein one or more of the thinning elements comprises a seam.
- 15.** A wearable seating assembly according to claim 14 wherein one or more of the thinning elements comprises sewing through the seat.
- 16.** A wearable seating assembly according to claim 1 wherein the seat further comprises a tapered region.
- 17.** A method of using a wearable seating assembly comprising: a. wearing a wearable seating assembly according to any one of claim 1 around a user's hips; b. in an upright position, allowing the first face of the wearable seating assembly to abut against the user's lower back and buttocks in an upright configuration; c. sitting on an external surface, allowing the seat to deform from an original shape such that a lower portion of the seat bends with respect to an upper portion of the seat to provide a seating surface between the user and the external surface in a seated configuration, without manually manipulating the wearable seat; and d. standing up from the external surface, allowing the seat to return to the original shape and the wearable seating assembly to the upright configuration, without manually manipulating the wearable seat.
- 18.** A method of using a wearable seating assembly comprising: a. wearing a wearable seating assembly according to claim 7 around a user's hips; and b. inserting the heating element through the heating element slot into the at least one heating element pocket.
- 19.** A method of using a wearable seating assembly according to claim 18, further comprising, when the at least one heating element pocket does not hold the heating element, maintaining a shape of the at least one heating element pocket by providing therein a placeholder material shaped like the heating element.
- 20.** A method of using a wearable seating assembly comprising: a. providing a wearable seating assembly according to claim 6; b. selecting between the first configuration and the second configuration by either: (i) rotating the seat approximately 180 degrees around a central longitudinal axis of the seat; or (ii) removing the strap from the seat and replacing the strap in a desired one of the first and second configurations.
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